

Western Mathematics

TOPIC REVISION QUESTIONS

Mathematics Standard 2

MS-F4 Investments and Loans

MS-F4.1: Investments

MS-F4.2: Depreciation and loans

Section I – Multiple Choice Questions (1 Mark each)

Section II – Longer Questions (Marks indicated beside questions)

Section I**Multiple Choice Questions (1 Mark each)****15 marks**

1. Anthony invests \$2 450 for 3 years in an account that pays 5% p.a. interest compounded annually.

How much interest does he earn?

- (A) \$367.50
 - (B) \$386.18
 - (C) \$390.22
 - (D) \$2 836.18
2. Lenore invests \$12 600 in an account that pays 4.8% p.a. interest compounded monthly. What is the value of her investment after $2\frac{1}{2}$ years?
- (A) \$13 124.00
 - (B) \$14 112.00
 - (C) \$14 166.86
 - (D) \$14 203.06
3. Sue and Jack wish to have \$20 000 for their granddaughter to start university in 6 years' time. What single sum they must invest now at 8% pa compounded annually to achieve their goal?
- (A) \$12 603
 - (B) \$13 200
 - (C) \$13 514
 - (D) \$96 000

4. Marcus wants to invest \$1 500.00 for 2 years. He can choose from the three accounts below.

Account X : Pays 5.2% pa simple interest.

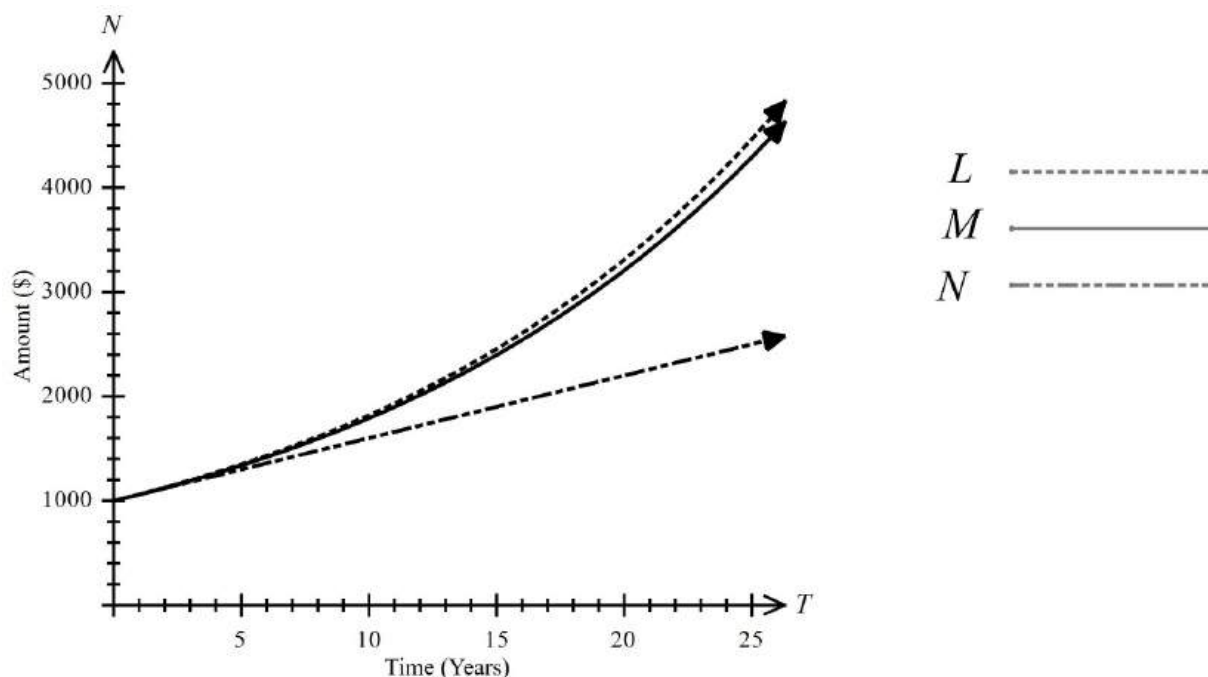
Account Y : Pays 5% pa interest, compounded annually.

Account Z : Pays 4.8% pa interest compounded monthly.

Which will give him the better return on his investment?

- (A) Account X
 - (B) Account Y
 - (C) Account Z
 - (D) Accounts X and Y are equal.
5. How much must be invested now (present value) invested at 8% p.a. compounding quarterly for 5 years if the future value is \$20 000.
- (A) \$4 290.96
 - (B) \$29 718.95
 - (C) \$13611.66
 - (D) \$13 459.43

6. Three friends, Leo, Marissa and Naomi invest \$1 000 in different accounts.
The graph below shows how their three investments would perform over 25 years.



The investments were (not in the same order as the names):

1. Simple interest of 6% pa.
2. Compound interest of 6% compounded monthly.
3. Compound interest of 6% compounded annually.

Which friend had investment number 2?

- (A) Leo
- (B) Marissa
- (C) Naomi
- (D) There isn't enough information to decide.
7. Joel invested a present value of \$2 0000 4 years ago and it grew to \$2 436.81 over 4 years. compounding half-yearly.

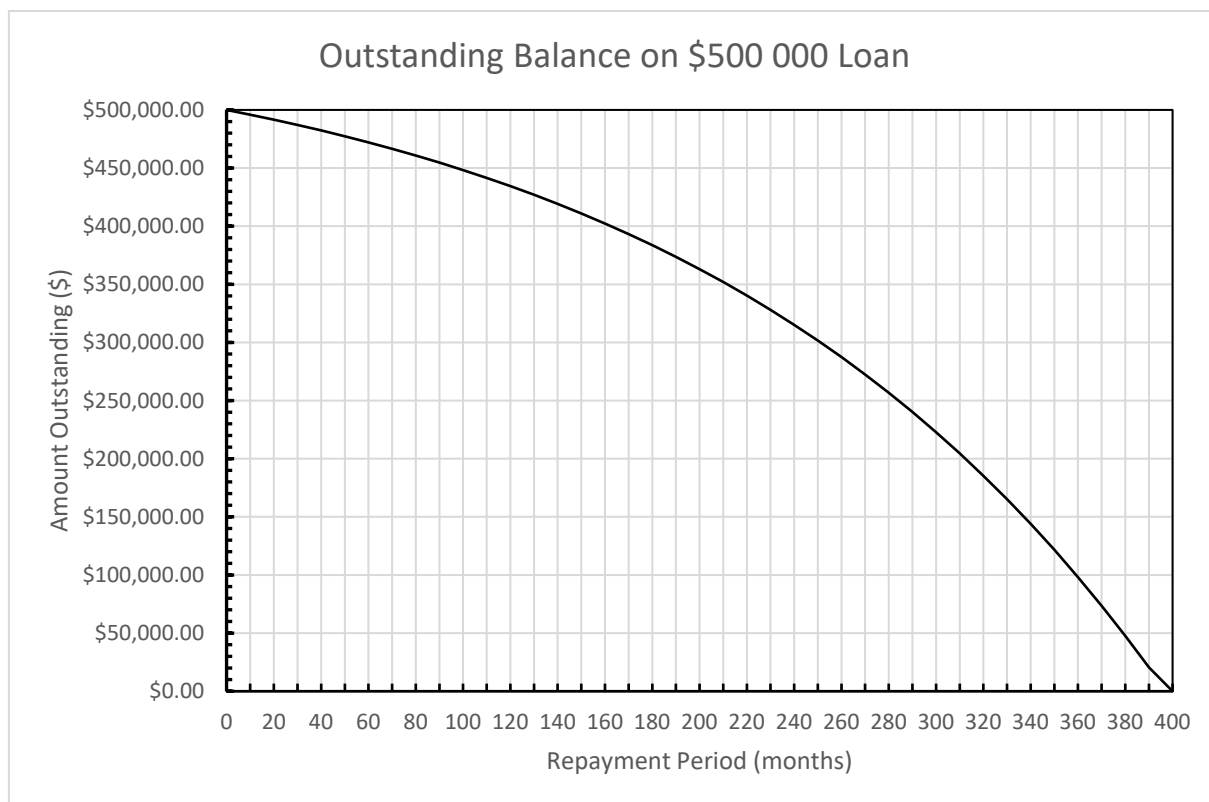
At what annual interest rate was the investment made?

- (A) 1.5% p.a.
- (B) 2.5% p.a.
- (C) 3.0% p.a.
- (D) 5.0% p.a.

8. Louisa has \$20 000 to invest for a period of three years. She has decided to invest the money in a bank account. She has narrowed her investment options to four options, all of which have no bank fees.

Which investment will earn her the most in interest?

- (A) Bank A offers 6.6% p.a. simple interest.
- (B) Bank B offers 6.5% p.a. interest compounding annually.
- (C) Bank C offers 6.4% p.a. interest compounding quarterly.
- (D) Bank D offers 6.0% p.a. interest compounding monthly.
9. The graph below shows the amount outstanding on a loan of \$500 000 with a term of 400 months.



After how many months has half of the amount borrowed been paid back?

- (A) 200 months
- (B) 280 months
- (C) 281 months
- (D) 284 months

10. Lily and Ethan start paying off a home loan for \$250 000 with an initial interest rate of 6% pa, calculated monthly. Lily creates a spreadsheet to keep track of their balance. She entered all interest rate changes and any changes they made to their repayment amount when they occurred.

Balance End

	A	B	C	D	E	F
2			Balance Start		Balance Start	
3	Date	Repayment	Month	Interest	Month	
4	Jan-14	\$1,600.00	\$250,000.00	\$1,250.00	\$249,650.00	
5	Feb-14	\$1,600.00	\$249,650.00	\$1,248.25	\$249,298.25	
6	Mar-14	\$1,600.00	\$249,298.25	\$1,246.49	\$248,944.74	
7	Apr-14	\$1,600.00	\$248,944.74	\$1,244.72	\$248,589.46	
8	May-14	\$2,000.00	\$248,589.46	\$1,242.95	\$247,832.41	
9	Jun-14	\$2,000.00	\$247,832.41	\$1,342.43	\$247,174.84	
10	Jul-14	\$2,000.00	\$247,174.84	\$1,338.86	\$246,513.70	
11	Aug-14	\$2,000.00	\$246,513.70	\$1,335.28	\$245,848.98	
12						
13						

Sheet1 Sheet2 Sheet3

In which month and by how much did the interest rate change?

- (A) It decreased by 1.0% pa in May.
- (B) It increased by 0.5% pa in June.
- (C) It increased by 1.0% pa in June.
- (D) It increased by 0.5% pa in July.

11. Mason borrows \$10 000 at 6% pa to buy a car and chooses to repay it in monthly repayments over 5 years.

Use the table below to find how much will he will pay for the car altogether.

Monthly Repayments on a loan of \$10 000

Time (years)

		2	3	4	5	6	7	8
Rate (% pa)	4	\$434.25	\$295.24	\$225.79	\$184.17	\$156.45	\$136.69	\$121.89
	5	\$438.71	\$299.71	\$230.29	\$188.71	\$161.05	\$141.34	\$126.60
	6	\$443.21	\$304.22	\$234.85	\$193.33	\$165.73	\$146.09	\$131.41
	7	\$447.73	\$308.77	\$239.46	\$198.01	\$170.49	\$150.93	\$136.34
	8	\$452.27	\$313.36	\$244.13	\$202.76	\$175.33	\$155.86	\$141.37

- (A) \$193.33
- (B) \$966.65
- (C) \$1599.80
- (D) \$11599.80
12. Ashleigh has a credit card which charges 18.5% pa simple interest calculated daily, which is debited to the card on the last day of each month. She had \$2 500 owing on her credit card on the 30th June after the interest was debited. She decides to pay off the total owing on the card before she makes any further purchases. She pays \$1 200 off the card on 30th June and pays off the remaining balance including interest on the 31st July.

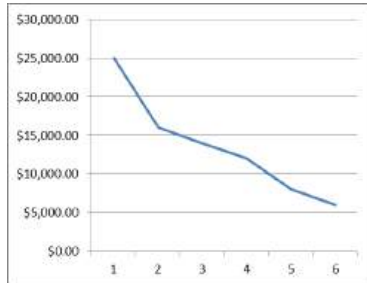
How much was paid on 31st July?

- (A) \$1 320.43
- (B) \$1 338.54
- (C) \$1540.50
- (D) \$3 342.60

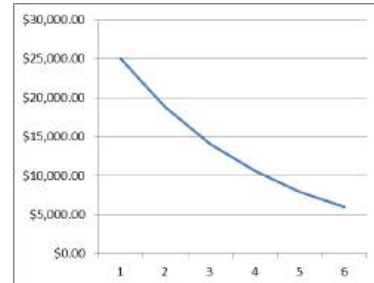
13. Caleb is investigating the different forms of depreciation of a vehicle. He collects the depreciation tables from four car dealers and graphs the value of a car initially valued at \$25 000 over 5 years under each table.

Which graph shows that a declining balance method of depreciation was used?

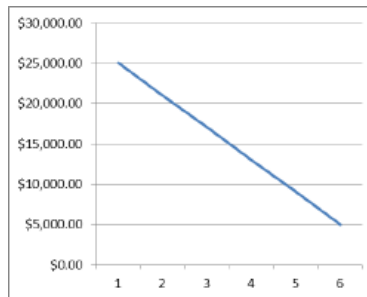
(A)



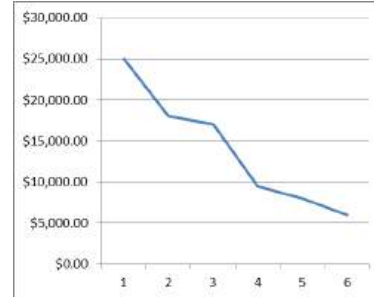
(B)



(C)



(D)



14. On the 1st January 2012 Rebecca invested \$1 450 in an account which pays 12% pa interest, compounded quarterly. On the 1st January 2013 she added a further \$1 280 to the account. How much has she in the account on the 1st January 2014?

- (A) \$3252.48
(B) \$2 911.99
(C) \$3 277.47
(D) \$5 604.25

Western Mathematics
Mathematics Standard 2
MS-F4 Investments and Loans
REVISION QUESTIONS - Section II

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.

Question 15 (3 marks)

Marks

Poppy bought a TV and paid a deposit of 20% of the cash price of \$1750, and monthly repayments of \$95.70 per month for 2 years.

(a) How much extra did she pay compared to paying cash for the TV?

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(b) What flat rate of interest per annum was she charged?

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Question 16 (3 marks)

Heather buys 500 shares in the Micro Guard company at a market value of \$4.50 per share.

- (a) If there are fixed brokerage fees of \$25.60 and stamp duty of 15 cents per \$100 or part thereof, calculate the total cost for Heather to buy the shares. 2

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- (b) Micro Guard pays a dividend of 60 cents per share. Calculate the percentage yield on the shares. 1

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Question 17 (2 marks)

Ebony has a credit card with no interest free period and an interest rate of 14% p.a. calculated daily. She had paid her card off in full before July and makes the following purchases during the month of July.

2 nd July	Cookpot	\$68.00
16 th July	DVDs	\$53.50
23 rd July	Petrol	\$40.00
26 th July	Makeup	\$105.50
29 th July	Dress	\$150.80

- Ebony pays her credit card bill in full on the 5th of August. How much interest does she pay? 2

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Question 18 (5 marks)

Jock depreciates his computer set-up using the declining balance method.

After substituting the interest rate into the formula, he obtains the equation $S = V_0 \times 0.87^n$.

- (a) What was the annual percentage rate at which the computer is depreciating? **1**

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- (b) What is the value of the computer after 3 years, if it was initially valued at \$2 499? **1**

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- (c) What percentage of the initial value remains after 5 years (to the nearest percent)? **1**

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- (d) Jock wishes to replace his computer every 4 years. Is this a good depreciation model for Jock to use? Justify your answer. **2**

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Question 19 (3 marks)

A computer system is purchased for \$5 800. Its depreciation can be calculated using a straight-line depreciation of \$1 200 p.a. or by using a declining balance rate of 30% p.a.

- (a) Find the value of the system after 6 years using the declining balance method.

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- (b) Which method will give the system the greater value after 3 years?

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Question 20 (2 marks)

An extract from a credit card statement is shown below.

2

Australian Commercial Bank			
Name : Nicholas Shoard		Account number 145 784 683	
Interest rate = 18% p.a.		Daily rate = 0.049%	
Date	Credit	Debit	Balance
01/03/15	Opening balance.		\$842.00
12/03/15		\$362 purchase	\$1 204.00
20/03/15		\$199 purchase	\$1 403.00
26/03/15	\$760 payment		\$643.00
31/03/15	Closing balance		\$643.00

If interest is calculated daily on the maximum daily balance and charged at the end of the month, calculate the interest that will be due for the month of March.

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Question 21 (4 marks)

- (a) Liza buys a car which has a market value of \$60 000 before on-road costs. 2
Stamp duty on the car is calculated at these rates:

- 3% of the market value up to and including \$45,000
- 5% of the market value over \$45,000.

Calculate the stamp duty payable on the purchase of the car.

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- (b) The car depreciates at a rate of 4.5% pa. 2
What is the value of the car 3 years after Liza buys it?

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Question 22 (7 marks)

Zane bought a car in July 2015 and wants to work out how long to keep the car.
 He asks for estimates of the resale value from three used car dealers.
 The results are summarised in the table below.

Dealer	Year							
	2015	2016	2017	2018	2019	2020	2021	2022
Davidson Auto	\$65,000	\$54,600	\$45,864	\$38,526	\$32,362	\$27,184	\$22,834	\$19,181
Gregson Motors	\$65,000	\$54,600	\$44,772	\$35,818	\$27,938	\$22,350	\$17,433	\$14,295
A1 Used Cars	\$65,000	\$57,000	\$49,000	\$41,000	\$33,000	\$25,000	\$17,000	\$9,000

- (a) Zane is deciding between keeping the car for either three years or six years.
 Outline how the result of this decision would influence his choice of dealer?

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- (b) A1 Used Cars uses the straight line depreciation method.

2

Explain how this method works and show how the straight line formula could be used to give the value of \$41 000 which appears in the table.

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- (c) A fourth dealer, Smiths Caryard, uses the declining balance method of depreciation with a depreciation rate of 14% pa.

1

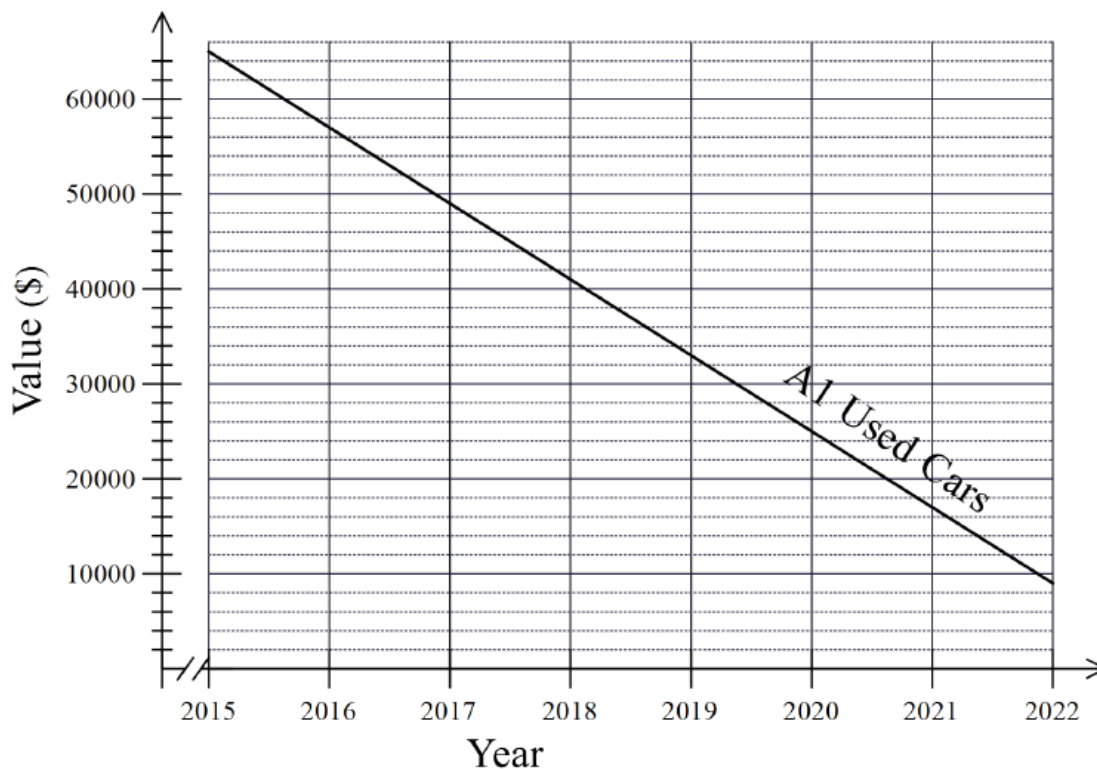
What is the value of a car according to Smiths after 5 years?

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- (d) The value of the car according to A1 Used Cars has been drawn on the graph below. On the same set of axes, draw a graph to show the value of the car according to Davidson's Autos for this period of time. 2



- (e) The value of the car according to Davidsons Autos and A1 Used Cars begins at the same value (\$65 000) in 2015. 1
Is the value according to both dealers ever equal again, and if so what is this value?

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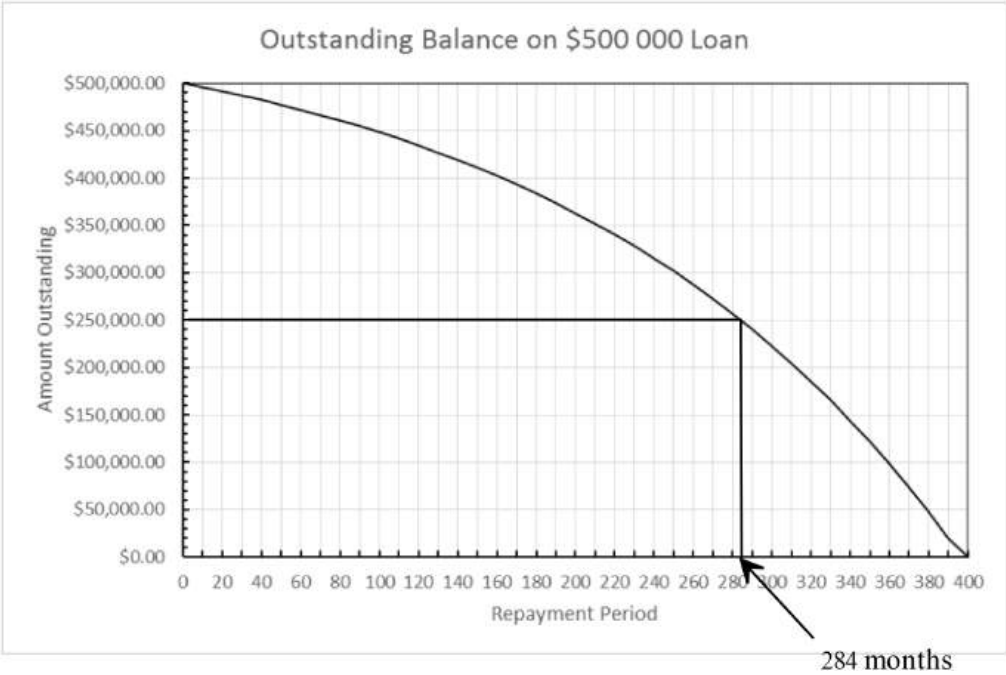
End of Revision

Western Mathematics MS-F4 Investments and Loans
Section I - Worked Solutions

Revision

No	Working	Answer
1	$FV = PV(1 + r)^n$ $= 2450 \times (1.05)^3$ $= \$2836.18125$ $\text{Interest} = 2836.18125 - 2450$ $= \$386.18$	B
2	$r = \frac{0.048}{12} = 0.004$ $n = 2.5 \times 12 = 30$ $FV = PV(1 + r)^n$ $= 12600 \times (1.004)^{30}$ $= \$14203.06025131$ $= \$14\,203.06$	D
3	$FV = PV(1 + r)^n$ $20000 = PV(1.08)^6$ $20000 = PV \times 1.58687432$ $PV = \frac{20000}{1.58687432}$ $= 12603.39256104$ $= \$12\,603 \text{ (nearest dollar)}$	A
4	Account X $PV = 1500, r = 0.052, n = 2$ $FV = PV + Prn$ $= 1500 + 1500 \times 0.052 \times 2$ $= \$1656.00$ Account Y $PV = 1500, r = 0.05, n = 2$ $FV = PV(1 + r)^n$ $= 1500(1.05)^2$ $= \$1653.75$ Account Z $PV = 1500, r = \frac{0.048}{12} = 0.004, n = 2 \times 12 = 24$ $FV = PV(1 + r)^n$ $= 1500(1.004)^{24}$ $= \$1650.82$ Account X is best for 2 years	A
5	$r = \frac{0.08}{4} = 0.02 \quad n = 5 \times 4 = 20$ $FV = PV(1 + r)^n$ $20000 = PV(1.02)^{20}$ $20000 = PV \times 1.4859474$ $PV = \frac{20000}{1.4859474} = 13459.42662573$ $= \$13\,459.43$	D

6	2. Compound interest of 6% compounded monthly. Number 1. would be the straight line. Number 2. and number 3. would be exponential curves, and number 2. would be steeper than number 3. because of the monthly compounding. So, matching them up: 2. L ----- 3. M ----- 1. N ----- So, the friend investment 2. is Leo.	A
7	$n = 4 \times 2 = 8$, $PV = 2000$, $FV = 2436.81$ $FV = PV(1 + r)^n$ $2436.81 = 2000(1 + r)^8$ $(1 + r)^8 = \frac{2436.81}{2000} = 1.218405$ Testing the values given A. $r = \frac{0.015}{2} = 0.0075 \Rightarrow (1.0075)^8 = 1.06159885$ NO B. $r = \frac{0.025}{2} = 0.0125 \Rightarrow (1.0125)^8 = 1.1044861$ NO C. $r = \frac{0.03}{2} = 0.015 \Rightarrow (1.015)^8 = 1.12649259$ NO D. $r = \frac{0.05}{2} = 0.025 \Rightarrow (1.025)^8 = 1.2184029$ YES	D
8	Bank A Interest = $20000 \times 0.066 \times 3 = 3960$ Value of Investment = \$23 960 Bank B Value of Investment = $20000 \times (1.065)^3 = \\$24\ 158.99$ Bank C Value of Investment = $20000 \times (1.016)^{12} = \\$24\ 196.61$ Bank D Value of Investment = $20000 \times (1.005)^{36} = \\$23\ 933.61$ Bank C gives the best interest.	C

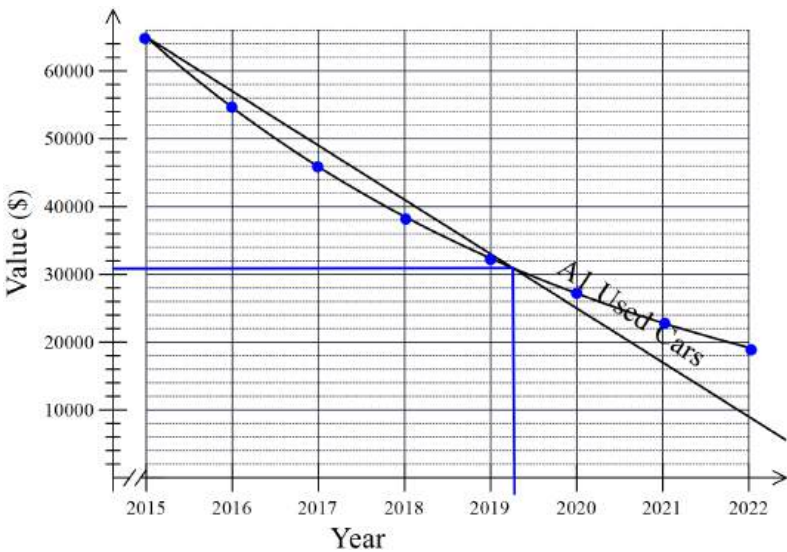
9		D
10	Looking down the interest column, the interest amount increased in June. Int Rate May = $\\$1,242.95 \div \\$248,589.46 = 0.0050$ (4 dec places) Int Rate June = $\\$1,342.43 \div \\$247,832.41 = 0.0054$ (4 dec places) Increase in annual interest rate = $0.0004 \times 12 \times 100 = 0.5\%$ (1 dec place)	B
11	Monthly repayment = \$193.33 from table. There are $5 \times 12 = 60$ repayments. Amount paid = $\\$193.33 \times 60 = \\$11\,599.80$	D
12	Amount owing for July = 1300. Daily interest rate = $\frac{18.5}{365} = 0.05\%$ per day Interest = $1300 \times 0.005 \times 31$ = \$20.43 Amount to Pay = $1300 + 20.43$ = \$1 320.43	A
13	Under the declining balance method, each succeeding years price is a percentage of the previous year's price, which means the value goes down by gradually lesser amounts each year. Graph B indicates this. Graph C is a straight line depreciation and A and D are erratic amounts.	B

14	Quarterly interest rate $= 0.12 \div 4 = 0.03$ After first year the investment $= 1450 \times (1.03)^4 = \\$1\,631.99$ After she added \$1 280 value $= \\$1\,631.99 + \\$1\,280 = \\$2\,911.99$ After second year the investment $= \\$2\,911.99 \times (1.03)^4 = \\$3\,277.47$	C
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MS-F4 Investments and Loans					Marks																																								
Section II Solution																																													
Question 15																																													
(a)	Deposit = 0.2×1750 = \$350.00 Payments = 24×\$95.70 = \$2 296.80 Total Paid = \$2 296.80 + \$350.00 = \$2 646.80 Interest = \$2 646.80 - 1750.00 = \$896.80				2																																								
(b)	Interest Rate = $\frac{\$896.80}{1400} \div 2$ = 32% p.a.				1																																								
Question 16																																													
(a)	Cost of shares = 500×\$4.50 = \$2 250.00 Stamp duty = 23×0.15 = \$3.45 Total cost = \$2 250 + \$3.45 + \$25.60 = \$2 279.05				2																																								
(b)	Yield = $\frac{60}{450} \times 100 = 13.3\%$				1																																								
Question 17																																													
(a)	<table><tr><td></td><td colspan="4">Rate per day = 0.14 ÷ 365 = 0.0003836</td></tr><tr><td>Amount</td><td>Days Interest</td><td>Total Amount</td><td colspan="2">Interest to 5th Aug</td></tr><tr><td>68.00</td><td>34</td><td>68.8925</td><td colspan="2">0.8925</td></tr><tr><td>53.50</td><td>20</td><td>53.9120</td><td colspan="2">0.4120</td></tr><tr><td>40.00</td><td>13</td><td>40.1999</td><td colspan="2">0.1999</td></tr><tr><td>105.50</td><td>10</td><td>105.9054</td><td colspan="2">0.4054</td></tr><tr><td>150.80</td><td>7</td><td>151.2054</td><td colspan="2">0.4054</td></tr><tr><td colspan="2">Total Interest = \$2.3152 = \$2.32</td><td colspan="3"></td></tr></table>					Rate per day = 0.14 ÷ 365 = 0.0003836				Amount	Days Interest	Total Amount	Interest to 5 th Aug		68.00	34	68.8925	0.8925		53.50	20	53.9120	0.4120		40.00	13	40.1999	0.1999		105.50	10	105.9054	0.4054		150.80	7	151.2054	0.4054		Total Interest = \$2.3152 = \$2.32					2
	Rate per day = 0.14 ÷ 365 = 0.0003836																																												
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Question 18																																													

(a)	$1 - r = 0.87$ $r = 1 - 0.87 = 0.13$ Rate is 13%	1
(b)	$S = 2499 \times 0.87^3$ $= 1645.598997$ $= \$1\,645.60$	1
(c)	$100 \times .87^5 = 49.842 \cong 50\%$ OR $S = 2499 \times 0.87^5$ $= 1245.55388083$ Percentage remaining $= \frac{1245.55388083}{2499} \times 100$ $= 49.84209207$ $= 50\%$	1
(d)	This model depreciates too slowly. After 4 years its value is $S = 2499 \times 0.87^4$ $= 1431.67112739$ Percentage remaining $= \frac{1431.67112739}{2499} \times 100$ $= 57.289761$ So only lost 43% of its value after 4 years and the fact that 50% remains after 5 years for computer equipment means the depreciation does not give sufficient loss in value to replace it after 4 years. Would need to use a different model, perhaps change the interest rate or use a straight line method.	2
Question 19		
(a)	$S = V_0(1 - r)^n$ $= 5800(1 - 0.3)^6$ $= \$682.36$	1
(b)	$S = V_0(1 - r)^n$ OR $S = V_0 - Dn$ $= 5800(1 - 0.3)^3$ $= 5800 - 1\,200 \times 3$ $= \$1989.40$ $= \$2\,200.00$ Straight line gives greater value	2

	Question 20 Interest from 1st to 11th = $842 \times 0.00049 \times 11 = 4.53838$ Interest from 12th to 19th = $1204 \times 0.00049 \times 8 = 4.71968$ Interest from 20th to 25th = $1403 \times 0.00049 \times 6 = 4.12482$ Interest from 26th to 31st = $643 \times 0.00049 \times 6 = 1.89042$ Total Interest = \$15.2733 = \$15.27	2
Question 21		
(a)	Stamp duty = 3% of 45 000 + 5% of 15 000 = $.03 \times 45\,000 + .05 \times 15\,000$ = $1\,350 + 750$ = \$2 100	1
(b)	$S = V_0(1 - r)^n$ = $60\,000(0.955)^3$ = \$52 259.03	1
Question 22		
(a)	If he keeps the car for 3 years A1 Used cars would give him the best resale price by around \$2 500 but after 6 year it would be the lowest, with Davidson's Auto being the best by around \$5 800.	1
(b)	The straight line depreciation method operates on the vehicle reducing by the same amount each year. In this case, the annual amount is $65000 - 57000 = \\$8\,000$. The formula $S = V_0 - Dn$ would become $S = 65000 - 8000n$ After three years $S = 65000 - 8000 \times 3 = \\$41\,000$ as in the table.	2
(c)	Use the formula $S = V_0(1 - r)^n$ $S = 65000(1 - 0.14)^5$ = $65000(0.86)^5$ = \$30578 (to the nearest dollar)	1

(d)	 In 2019 the value is about \$31 000 for both methods.	2
(e)	They both give a value of around \$31 000 late in 2019.	1